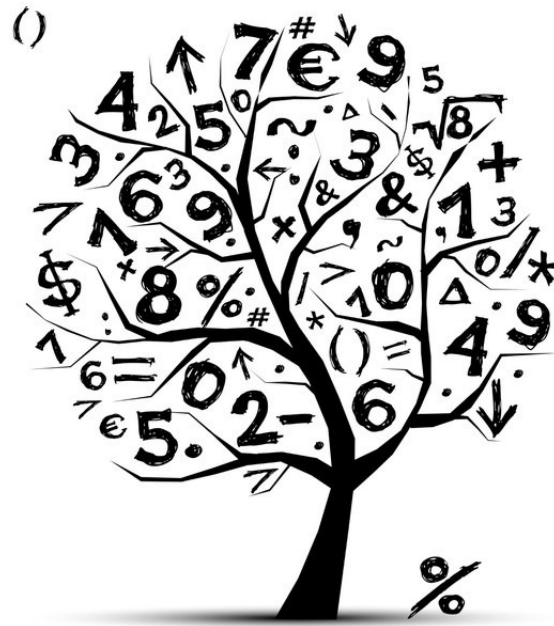


Last update: 7 ottobre 2026

System of linear equations



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1 Existence and uniqueness of solutions

Consider a system of m linear equations with n variables:

$$A \mathbf{x} = \mathbf{b}$$

where:

- $A \in \mathbb{R}^{m \times n}$ is the coefficient matrix (m equations, n variables)
- $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is the vector of variables
- $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is the right-hand side vector
- $(A|\mathbf{b}) \in \mathbb{R}^{m \times (n+1)}$ is the augmented matrix

The existence and uniqueness of solutions to the system $A \mathbf{x} = \mathbf{b}$ depend on the rank of the coefficient matrix A and the rank of the augmented matrix $(A|\mathbf{b})$.

Case 1 - No solution (inconsistent system):

$$\text{rank}(A) < \text{rank}(A|\mathbf{b})$$

The system is inconsistent and has no solution.

Case 2 - Unique solution:

$$\text{rank}(A) = \text{rank}(A|\mathbf{b}) = n$$

The system has a unique solution.

Case 3 - Infinite solutions:

$$\text{rank}(A) = \text{rank}(A|\mathbf{b}) < n$$

The system has infinitely many solutions with $n - \text{rank}(A)$ degrees of freedom (free variables).

2 Gaussian Elimination Method

The **Gaussian elimination method** is an algorithm for solving systems of linear equations by transforming the augmented matrix $(A|\mathbf{b})$ into **row echelon form** (upper triangular form) through elementary row operations.

- Given the system of linear equations:

$$A x = b$$

we form the **augmented matrix**:

$$(A|b)$$

- The method consists of two phases:

Phase 1 - Forward elimination: Transform the augmented matrix into upper triangular form using elementary row operations:

- Swap two rows
- Multiply a row by a non-zero scalar
- Add a multiple of one row to another row

The goal is to create zeros below the diagonal, obtaining:

$$\left(\begin{array}{cccc|c} \tilde{a}_{11} & \tilde{a}_{12} & \cdots & \tilde{a}_{1n} & \tilde{b}_1 \\ 0 & \tilde{a}_{22} & \cdots & \tilde{a}_{2n} & \tilde{b}_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \tilde{a}_{nn} & \tilde{b}_n \end{array} \right)$$

Phase 2 - Backward substitution: Solve the upper triangular system from bottom to top:

$$\begin{aligned} x_n &= \frac{\tilde{b}_n}{\tilde{a}_{nn}} \\ x_{n-1} &= \frac{\tilde{b}_{n-1} - \tilde{a}_{n-1,n} x_n}{\tilde{a}_{n-1,n-1}} \\ &\vdots \\ x_i &= \frac{\tilde{b}_i - \sum_{j=i+1}^n \tilde{a}_{ij} x_j}{\tilde{a}_{ii}} \end{aligned}$$

Elementary row operations:

- $R_i \leftrightarrow R_j$: swap row i with row j
- $R_i \leftarrow k R_i$: multiply row i by scalar $k \neq 0$
- $R_i \leftarrow R_i + k R_j$: add k times row j to row i

These operations do not change the solution set of the system.

Esempio 1: Solving a system using Gaussian elimination - Part 1

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

Augmented matrix:

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 4 & 3 & 3 & 10 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

Step 1: Eliminate x_1 from rows 2 and 3

$R_2 \leftarrow R_2 - 2R_1$ (subtract 2 times row 1 from row 2):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 8 & 7 & 9 & 24 \end{array} \right)$$

$R_3 \leftarrow R_3 - 4R_1$ (subtract 4 times row 1 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 3 & 5 & 8 \end{array} \right)$$

Step 2: Eliminate x_2 from row 3

$R_3 \leftarrow R_3 - 3R_2$ (subtract 3 times row 2 from row 3):

$$\left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \end{array} \right)$$

Upper triangular form achieved! The corresponding system is:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 1x_2 + 1x_3 = 2 \\ 2x_3 = 2 \end{cases}$$

Esempio 2: Solving a system using Gaussian elimination - Part 2**Backward substitution:**

From the third equation:

$$2x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - x_3 = 2 - 1 = 1$$

From the first equation:

$$2x_1 + x_2 + x_3 = 4 \implies 2x_1 = 4 - x_2 - x_3 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Solution:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = \mathbf{b} \quad \checkmark$$

3 LU Factorization Method

The **LU factorization** (or **LU decomposition**) is a method for solving systems of linear equations by decomposing the coefficient matrix A into the product of two triangular matrices:

$$A = LU$$

where L is a **lower triangular matrix** (with 1's on the diagonal) and U is an **upper triangular matrix**.

- Given the system of linear equations:

$$Ax = b$$

if we have the LU factorization $A = LU$, we can substitute:

$$LUx = b$$

- We introduce the **auxiliary variable** y defined as:

$$y := Ux$$

so that $LUx = b$ becomes $Ly = b$.

Why does this work? We are splitting the original system $Ax = b$ into two simpler triangular systems:

- Find y such that $Ly = b$.
- Find x such that $Ux = y$.

If both steps succeed, then $Ax = LUx = Ly = b$, so x is indeed a solution of the original system. Since both L and U are triangular, each of the two systems can be solved in $O(n^2)$ operations by simple substitution.

- We then solve the system in two steps:

Step 1 - Forward substitution: Solve $Ly = b$ for y

Since L is lower triangular, we can solve this system easily from top to bottom:

$$\begin{aligned} y_1 &= b_1 \\ y_2 &= b_2 - \ell_{21} y_1 \\ y_3 &= b_3 - \ell_{31} y_1 - \ell_{32} y_2 \\ &\vdots \\ y_i &= b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j \end{aligned}$$

Step 2 - Backward substitution: Solve $\mathbf{U} \mathbf{x} = \mathbf{y}$ for $\mathbf{x}$

Since $\mathbf{U}$ is upper triangular, we can solve this system easily from bottom to top:

$$\begin{aligned} x_n &= \frac{y_n}{u_{nn}} \\ x_{n-1} &= \frac{y_{n-1} - u_{n-1,n} x_n}{u_{n-1,n-1}} \\ &\vdots \\ x_i &= \frac{y_i - \sum_{j=i+1}^n u_{ij} x_j}{u_{ii}} \end{aligned}$$

Advantages of LU factorization:

- Once the factorization $\mathbf{A} = \mathbf{L} \mathbf{U}$ is computed, we can solve the system for different right-hand sides $\mathbf{b}$ efficiently
- Both forward and backward substitution require only $O(n^2)$ operations
- The factorization itself requires $O(n^3)$ operations, but it needs to be done only once

Esempio 3: Solving a system using LU factorization - Part 1

Consider the system of linear equations:

$$\begin{cases} 2x_1 + 1x_2 + 1x_3 = 4 \\ 4x_1 + 3x_2 + 3x_3 = 10 \\ 8x_1 + 7x_2 + 9x_3 = 24 \end{cases}$$

In matrix form: $\mathbf{A} \mathbf{x} = \mathbf{b}$ where

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

Given the LU factorization of $\mathbf{A}$:

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \quad \mathbf{U} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Verification:

$$\mathbf{L} \mathbf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} = \mathbf{A} \quad \checkmark$$

Esempio 4: Solving a system using LU factorization - Part 2

Step 1 - Forward substitution: Solve $L y = b$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix}$$

From the first equation:

$$y_1 = 4$$

From the second equation:

$$2 y_1 + y_2 = 10 \implies y_2 = 10 - 2 \cdot 4 = 10 - 8 = 2$$

From the third equation:

$$4 y_1 + 3 y_2 + y_3 = 24 \implies y_3 = 24 - 4 \cdot 4 - 3 \cdot 2 = 24 - 16 - 6 = 2$$

Thus:

$$y = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

Step 2 - Backward substitution: Solve $U x = y$

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

From the third equation:

$$2 x_3 = 2 \implies x_3 = 1$$

From the second equation:

$$x_2 + x_3 = 2 \implies x_2 = 2 - 1 = 1$$

From the first equation:

$$2 x_1 + x_2 + x_3 = 4 \implies 2 x_1 = 4 - 1 - 1 = 2 \implies x_1 = 1$$

Solution:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verification:

$$A x = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2+1+1 \\ 4+3+3 \\ 8+7+9 \end{pmatrix} = \begin{pmatrix} 4 \\ 10 \\ 24 \end{pmatrix} = b \quad \checkmark$$

4 Cramer's rule

Cramer's rule is an explicit formula for the solution of a system of m linear equations with m variables (valid whenever the system has a unique solution).

- Given a column vector $\mathbf{b} \in \mathbb{R}^{m \times 1}$ of m rows, a matrix $\mathbf{A} \in \mathbb{R}^{m \times m}$ of m rows and m columns and a column vector $\mathbf{x} \in \mathbb{R}^{m \times 1}$ of m rows containing the m variables:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mm} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

if $\det(\mathbf{A}) \neq 0$, the solution $\tilde{\mathbf{x}}$ of the system of m linear equations:

$$\mathbf{A} \mathbf{x} = \mathbf{b}$$

is given by the formula:

$$\tilde{x}_j = \frac{\det(\mathbf{A}_j)}{\det(\mathbf{A})}, \quad \forall j \in \{1, 2, \dots, m\} \quad (1)$$

where $\mathbf{A}_j$ is the matrix formed by replacing the j -th column of $\mathbf{A}$ by the column vector $\mathbf{b}$.

4.1 Systems of two equations and two variables

With $m = 2$, we have:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \quad \mathbf{A}_1 = \begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = a_{11} a_{22} - a_{12} a_{21}$$

$$\det(\mathbf{A}_1) = b_1 a_{22} - a_{12} b_2$$

$$\det(\mathbf{A}_2) = a_{11} b_2 - b_1 a_{21}$$

If $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} a_{11} x_1 + a_{12} x_2 = b_1 \\ a_{21} x_1 + a_{22} x_2 = b_2 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{\det(\mathbf{A}_1)}{\det(\mathbf{A})}, \frac{\det(\mathbf{A}_2)}{\det(\mathbf{A})} \right)$$

Esempio 5: solution of a systems of two equations and two variables

- Given

$$\mathbf{A} = \begin{pmatrix} -1 & 1 \\ 8 & 2 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ 19 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} 2 & 1 \\ 19 & 2 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & 2 \\ 8 & 19 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = (-1) \cdot 2 - 1 \cdot 8 = -10$$

$$\det(\mathbf{A}_1) = 2 \cdot 2 - 1 \cdot 19 = -15$$

$$\det(\mathbf{A}_2) = (-1) \cdot 19 - 2 \cdot 8 = -35$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 + x_2 = 2 \\ 8x_1 + 2x_2 = 19 \end{cases} \implies (x_1, x_2) = \left(\frac{-15}{-10}, \frac{-35}{-10} \right) = \left(\frac{3}{2}, \frac{7}{2} \right)$$

- Given

$$\mathbf{A} = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \end{pmatrix}$$

we have

$$\mathbf{A}_1 = \begin{pmatrix} -2 & -1 \\ 0 & -1 \end{pmatrix} \quad \mathbf{A}_2 = \begin{pmatrix} -1 & -2 \\ 1 & 0 \end{pmatrix}$$

and

$$\det(\mathbf{A}) = (-1) \cdot (-1) - (-1) \cdot 1 = 2$$

$$\det(\mathbf{A}_1) = (-2) \cdot (-1) - (-1) \cdot 0 = 2$$

$$\det(\mathbf{A}_2) = (-1) \cdot 0 - (-2) \cdot 1 = 2$$

Since $\det(\mathbf{A}) \neq 0$, we then have:

$$\begin{cases} -x_1 - x_2 = -2 \\ x_1 - x_2 = 0 \end{cases} \implies (\tilde{x}_1, \tilde{x}_2) = \left(\frac{2}{2}, \frac{2}{2} \right) = (1, 1)$$